Demo Study A: LSTM Prediction of Rain Attenuation on a Ka-band Satellite Link

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Abstract

This paper proposes an LSTM model to predict rain attenuation on a Ka-band satellite link. We evaluate the model on a single-site beacon dataset and show that it outperforms the ITU-R P.618 baseline in RMSE.

Introduction

Rain fade is the dominant impairment for Ka-band satellite communication. Accurate attenuation forecasts enable adaptive coding and modulation. This study aims to improve short-term forecasting using recurrent neural networks.

Data

The dataset consists of beacon measurements collected over 18 months at a single ground station, sampled every 1 second. Rain gauge records were collected in parallel and aligned with the beacon signal. The data were split chronologically into training and test sets.

Methodology

We use a two-layer LSTM with 64 hidden units, trained with the Adam optimizer. Input features include rain rate, elevation angle, and past attenuation values. The model is compared with the ITU-R P.618 empirical model and a linear regression baseline.

Results

The LSTM achieves an RMSE of 0.82 dB, which improves on the ITU-R P.618 baseline RMSE of 1.54 dB. Correlation coefficient reaches 0.93 on the test set. Performance degrades for attenuation events above 15 dB where training samples are scarce.

Limitations

The evaluation is limited to a single site, and generalization to other locations was not evaluated. Peak attenuation events are rare in the dataset, so event-level errors remain high.